Legal Case Retrieval: An RAG-Based Legal Assistant

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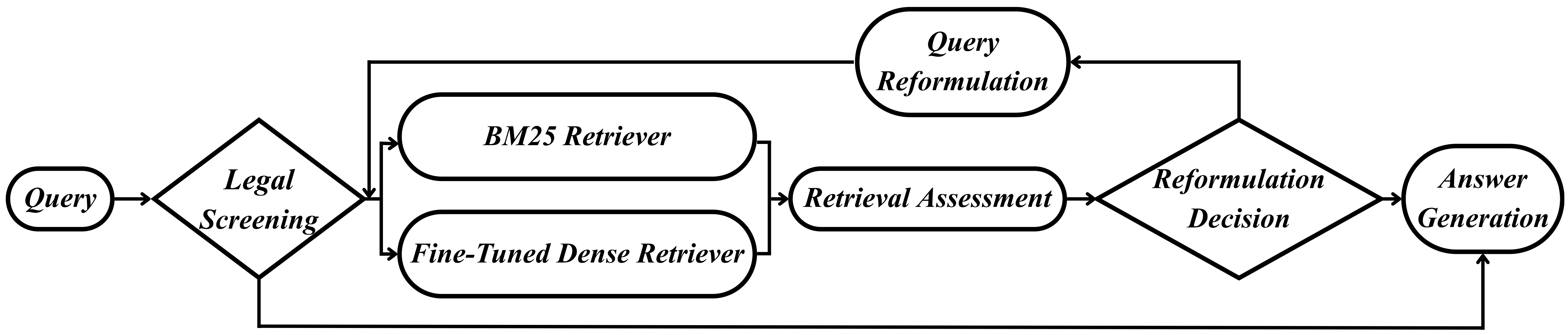
Abstract

Traditional legal case retrieval relies heavily on keyword matching, which often struggles with semantic variation and complex legal queries. This project proposes a ***retrieval-augmented legal question-answering system*** that combines ***BM25***, a ***fine-tuned dense retriever***, and an ***agentic multi-step workflow*** for query reformulation and evidence assessment. Experimental results show improvements in ***correctness*** and ***faithfulness*** over both LLM-only and baseline RAG settings. The project demonstrates how hybrid retrieval and evidence-grounded generation can improve reliability in domain-specific question answering.

Architecture of the Proposed RAG Framework



Methodology

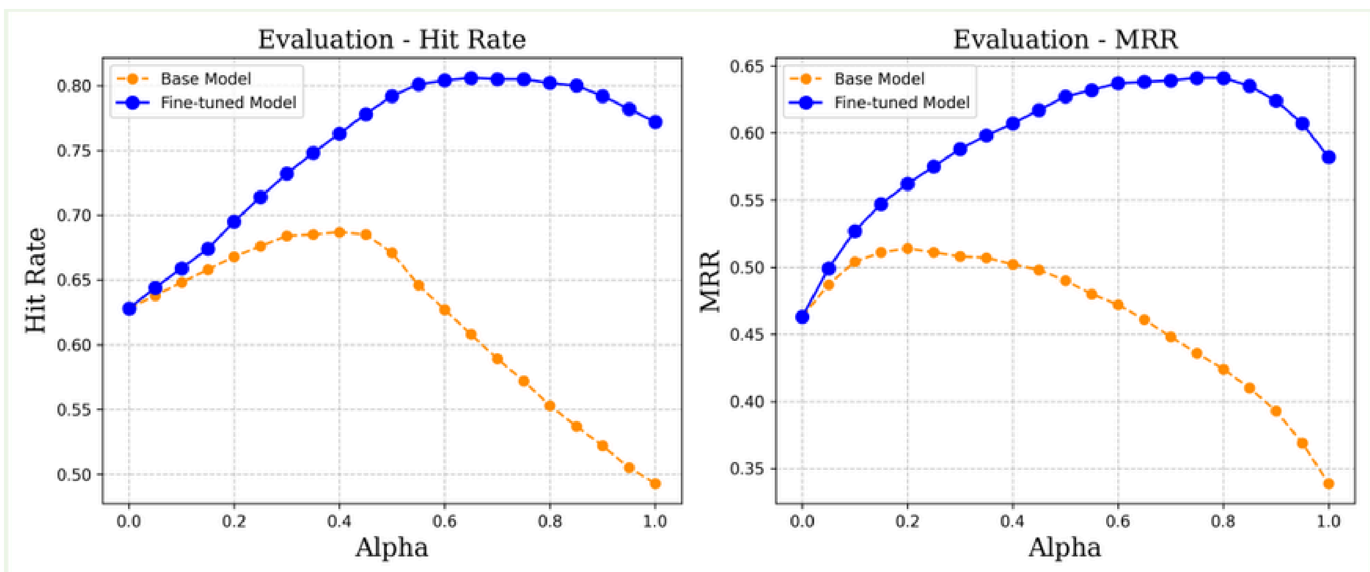
- **Data Collection and Q&A Construction:** Judicial Yuan rulings were collected as the legal corpus. An LLM was used to generate a synthetic ***Q&A dataset*** for retriever fine-tuning and downstream evaluation.

- **Dense Retriever Fine-Tuning:** A multilingual embedding model was fine-tuned on the Q&A pairs using ***contrastive learning,*** improving ***HitRate@10*** by ***0.23*** over the base model.

- **Hybrid Retrieval and Agentic RAG:** The system combined ***BM25*** with the fine-tuned dense retriever in ***Weaviate***. Built with ***LangGraph***, the workflow assessed retrieval adequacy, reformulated queries when necessary, and generated answers from accumulated evidence.

- **Evaluation:** System performance was assessed using ***correctness*** and ***faithfulness***. The proposed agentic RAG framework achieved the strongest overall performance among the compared settings.

Results and Evaluation



System Configuration	Correctness	Faithfulness
LLM-Only	0.86	N/A
Naïve RAG (Baseline)	0.96	0.89
Agentic RAG (Proposed)	1.00	0.99

Alignment with the SDGs:

SUSTAINABLE DEVELOPMENT GOALS

16 PEACE, JUSTICE AND STRONG INSTITUTIONS

Relevance to Intelligent Robotics

This project strengthened my ability to design intelligent systems that integrate **information retrieval, iterative reasoning, and evidence-based decision-making**, with an emphasis on **reliability, uncertainty handling, and adaptive workflows**.